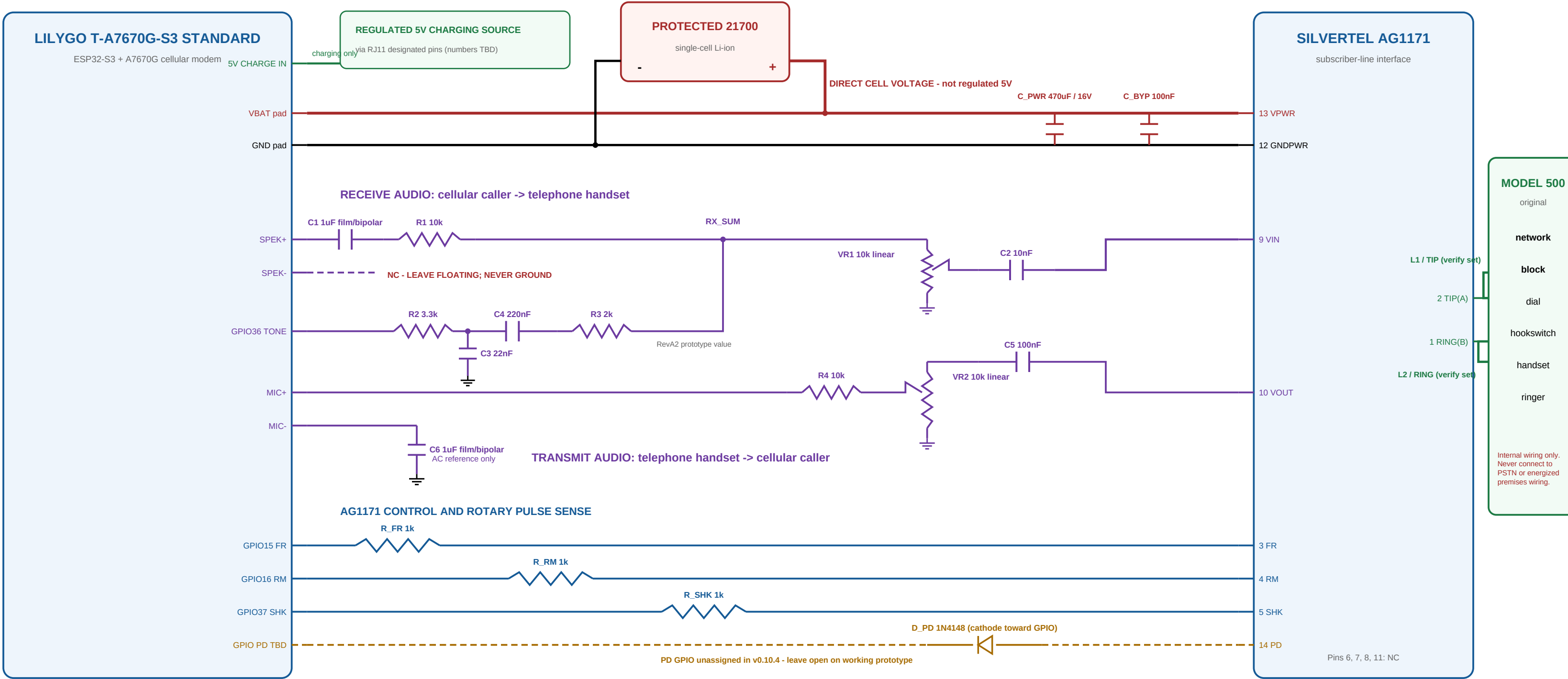


RotaryCell - Complete Prototype Wiring and Component Schematic

Point-to-point hand-wiring guide for the working transformerless prototype

LEGEND: battery power audio control common ground optional / not fitted No PCB/connector abstraction: wires terminate at module pins and components



BUILD / SAFETY NOTES

- Audio values are the working RevA2 prototype values; R3 is 2k.
- SPEK- stays floating. MIC- reaches ground only through the 1uF AC-reference capacitor.
- The protected 21700 directly feeds LilyGO VBAT and AG1171 VPWR. Never put charging 5V on VPWR.
- Set both 10k level trimmers for minimum signal before the first call, then raise slowly.
- AG1171 pins 6, 7, 8, 11 are unconnected. Tip/Ring are internal to the telephone only.
- RJ11 pin numbers and physical switch solder pads still require labeled photos.

IMPORTANT: HAND-WIRING FITS, BUT PACKAGING IS CHALLENGING

The prototype can be built entirely from the modules and discrete components shown here and installed inside the original telephone without modifying it. Point-to-point parts and wiring occupy substantially more space than the PCB version, so expect repeated dry-fitting and careful routing.

- Lay out the LilyGO, protected 21700, AG1171, trimmers and audio components before cutting final wire lengths.
- Insulate every exposed joint and provide strain relief. Keep wiring clear of the dial, hookswitch and other moving metalwork.
- Check that the base closes without pinching wires or loading terminals. A complete hand-wired installation is absolutely possible, but it requires patience and deliberate placement.